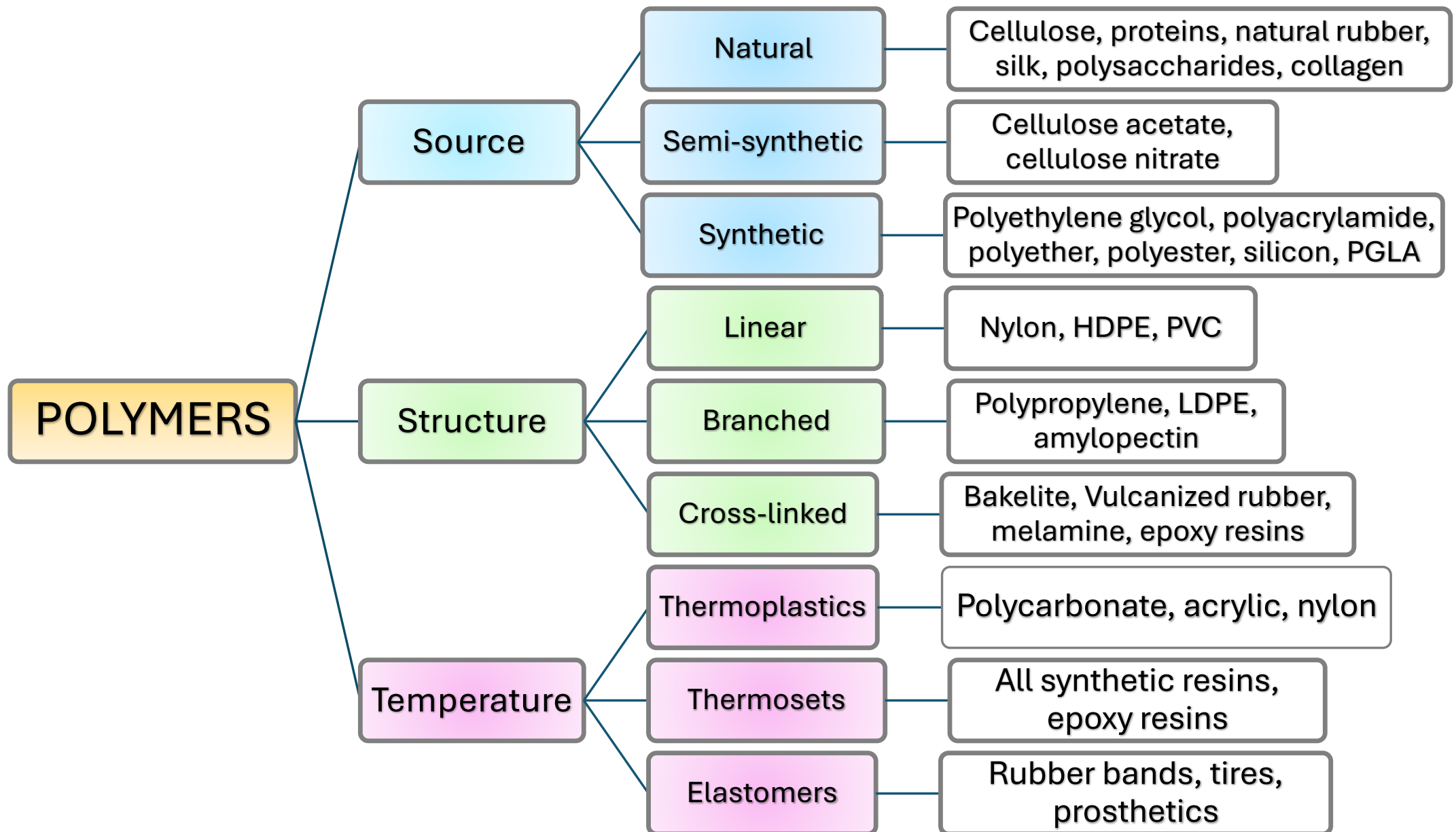


What are polymers?



Polymer curing is a chemical process that hardens and toughens polymers, usually from a liquid/gel to a solid, by creating cross-links between polymer chains, forming a rigid 3D network through reactions like polymerization or polycondensation

Thermoplastic material (or thermosoftening plastic) is a type of plastic polymer material (or plastomer) that becomes pliable at elevated temperatures and becomes less so once it has cooled. Thermoplastic materials can withstand very high temperatures so the heat required for melt processing must be even higher. Unlike thermosets, thermoplastic materials can be molded as often as is desired by cooling and reheating them, as long as the material is not overheated. Overheating causes the material to decompose. Interestingly, some thermoplastics can actually be welded.

Thermosets are plastics that retain their condition and shape after curing due to spatial crosslinking. Once cured, their polymers can no longer be dissolved. This type of plastic is popular in electrical installations due to its mechanical and chemical resistance, even at high temperatures. Popular thermosets include phenoplastics, virtually all synthetic resins (such as epoxy resins), polyurethane resins for paints and surface coatings and polyester.

Thermoplastics	Thermosets
No chemical curing reaction	Chemical curing reaction
High viscosity	Low viscosity
Difficult fiber impregnation	Good fiber impregnation
Conditionally solvent resistant	High solvent resistance
Short process times	Medium-to-long process times
Unlimited shelf or storage life	Limited shelf or storage life
Materials are weldable	High fixture effort
High energy absorption in case of damage	Brittle fracture behavior in case of damage
Good recycling properties	Limited recycling properties

THERMOPLASTIC MATERIALS

Polycarbonate
Polyoxymethylene
Acetal Homopolymer
Acrylic
Nylon
Polyethylene
Polypropylene
Polystyrene
Polyvinyl chloride
Teflon

VS.

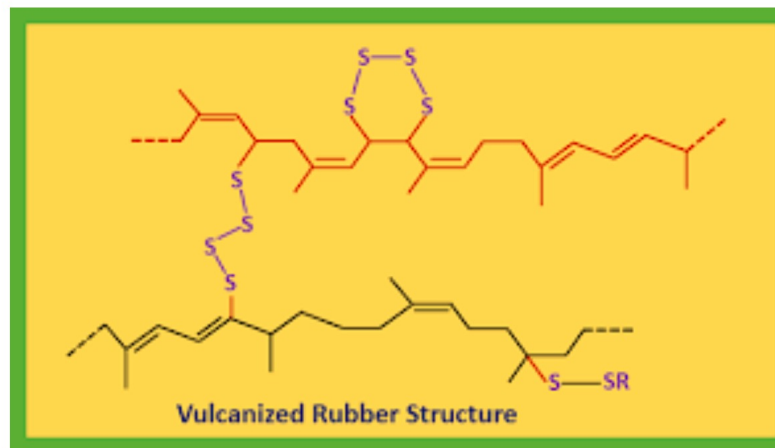
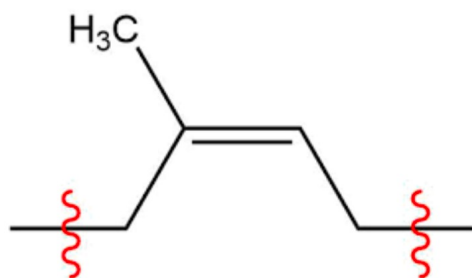
THERMOSET MATERIALS

Vulcanized rubber
Bakelite
Polyurethane
Epoxy resin
Vinyl ester resin
Polyester
Polyurea
Melamine
Silicone
Urea Formaldehyde

Elastomers (or “elastic polymers”) are polymers that are held together by weak intermolecular forces, generally exhibiting low Young’s modulus and high yield strength or high failure strain. They are also viscous as well as elastic, a property known as viscoelasticity. These plastics have unique properties that allow them to deform or stretch under extreme tensile and compressive loads then return to their original shape.

Examples of elastomers include natural rubber, polyurethanes, polybutadiene, silicone and neoprene. This type of plastic is popular as a material for tires, rubber bands, baby pacifiers, belts, prosthetics and other items of this nature.

Rubbers contain cis alkenes



Vulcanization is carried out by heating crude rubber in presence of sulfur or dipping it in a solution of S_2Cl_2 in CS_2 . Vulcanization depends upon the amount of sulfur used (by increasing the amount of sulfur the rubber can be hardened), temperature, and duration of heating

Vulcanization increases elasticity and ability to regain back shape

How to synthesize polymers?

Linking monomers

Addition polymerization
(chain-growth)

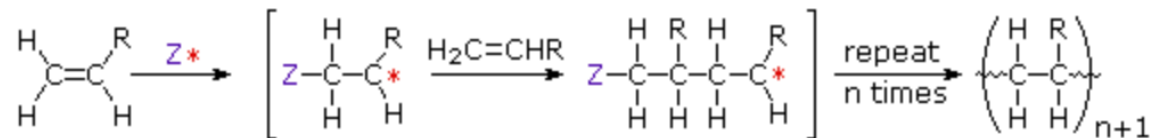
Condensation polymerization
(step-growth)

Polyvinyl chloride (PVC)

Polyethylene

Polystyrene

Polypropylene

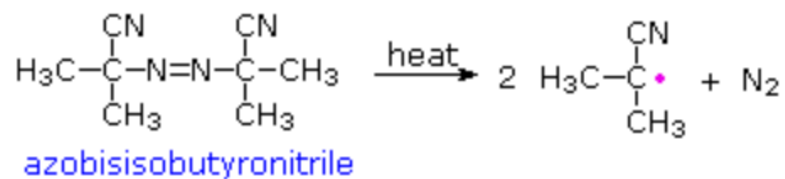
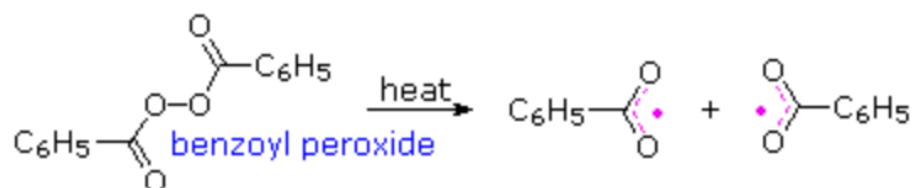
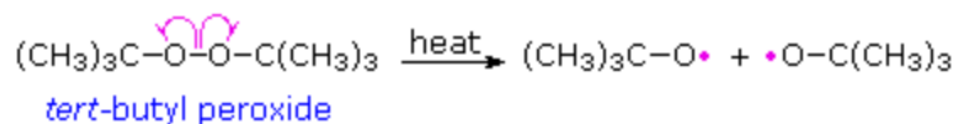


Z^* is an initiating species

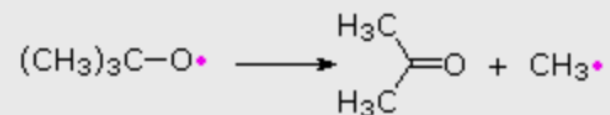
* may be a radical, a cation or an anion

- Radical Polymerization The initiator is a radical, and the propagating site of reactivity (*) is a carbon radical.
- Cationic Polymerization The initiator is an acid, and the propagating site of reactivity (*) is a carbocation.
- Anionic Polymerization The initiator is a nucleophile, and the propagating site of reactivity (*) is a carbanion.
- Coordination Catalytic Polymerization The initiator is a transition metal complex, and the propagating site of reactivity (*) is a terminal catalytic complex.

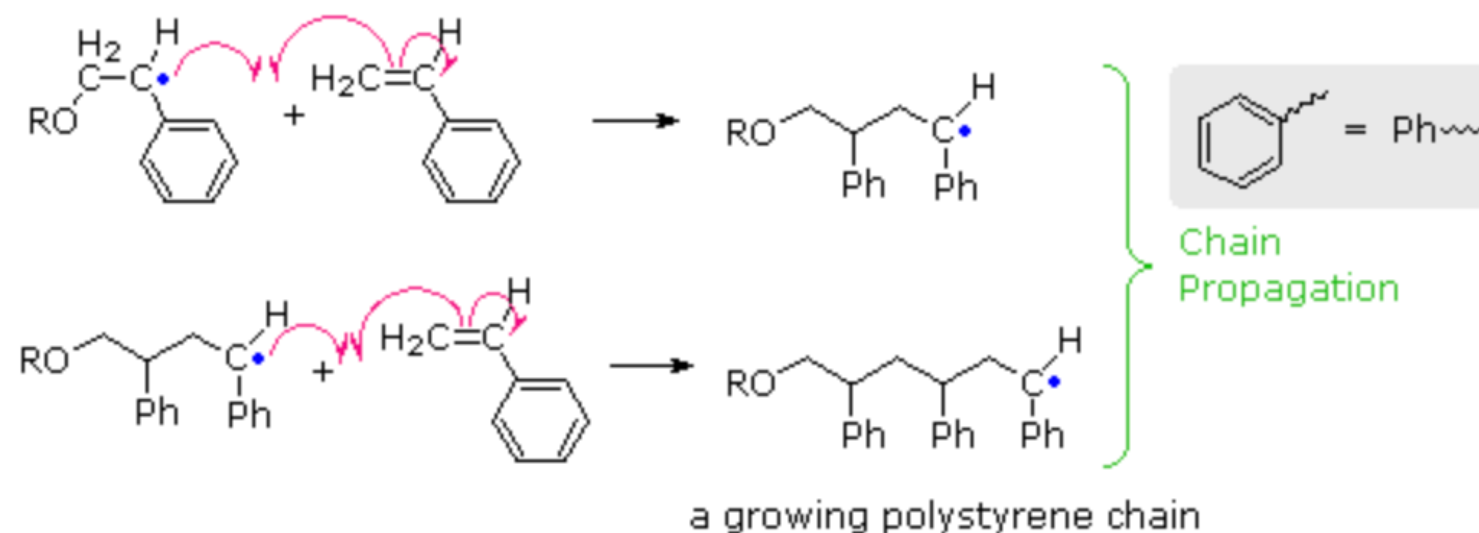
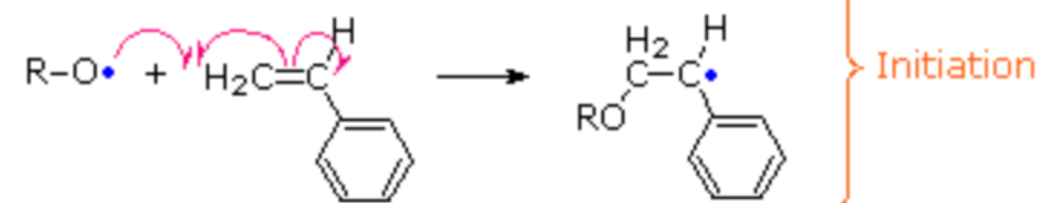
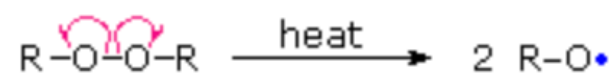
Some Radical Initiators



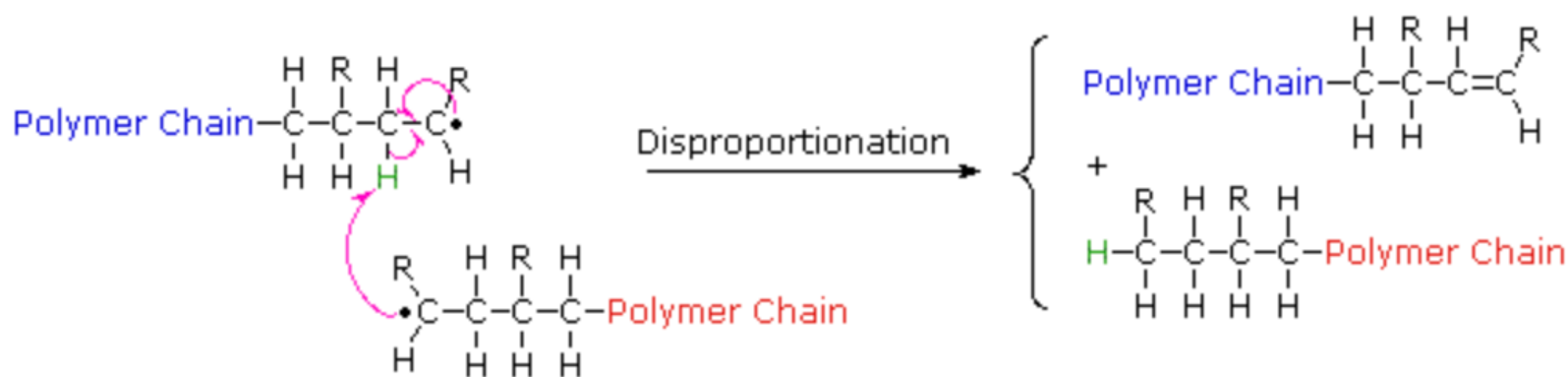
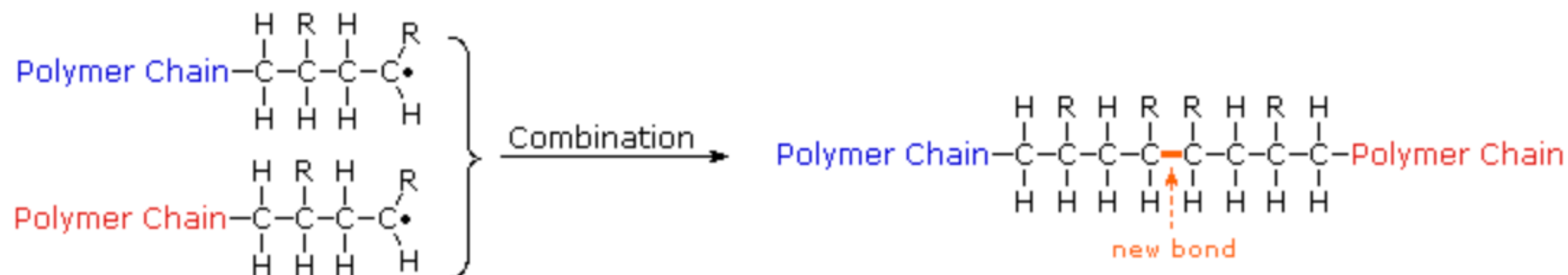
Subsequent Reactions



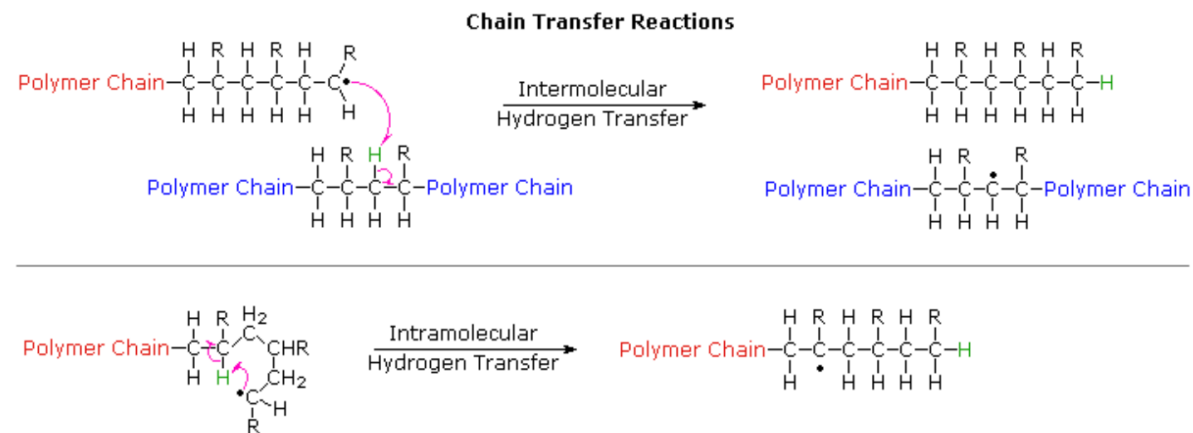
UV light/ visible light can also lead to radical formation



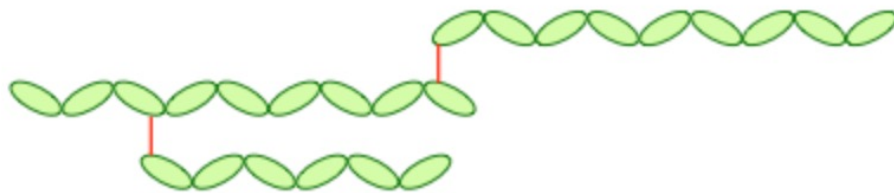
Chain Termination Reactions



Another reaction that diverts radical chain-growth polymerizations from producing linear macromolecules is called chain transfer. As the name implies, this reaction moves a carbon radical from one location to another by an intermolecular or intramolecular hydrogen atom transfer (colored green). These possibilities are demonstrated by the following equations



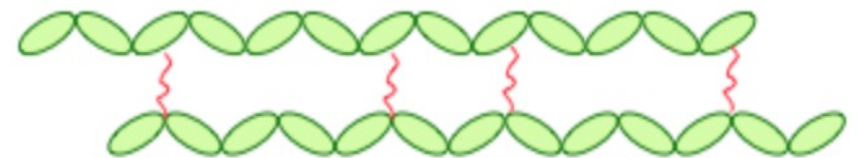
Chain topology



branched homopolymer

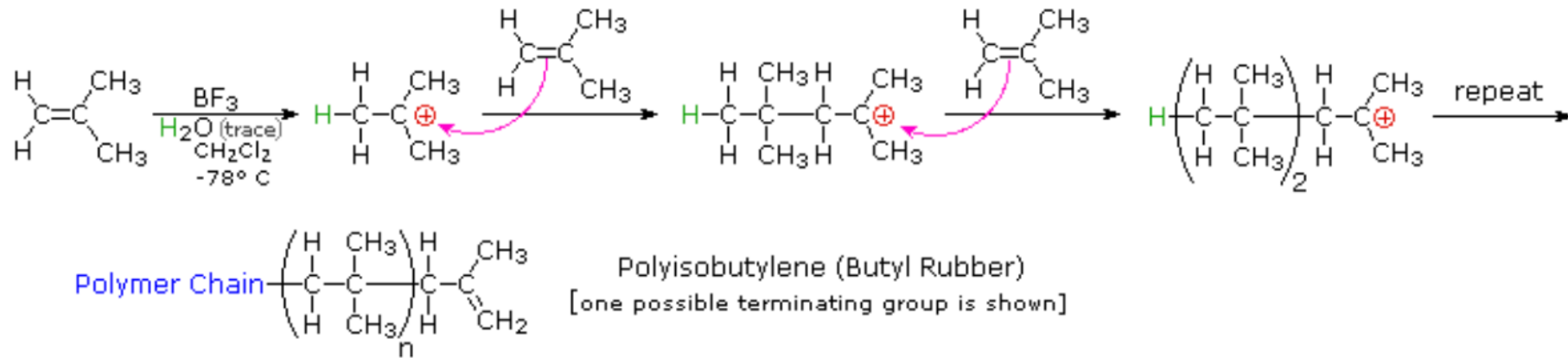


graft copolymer

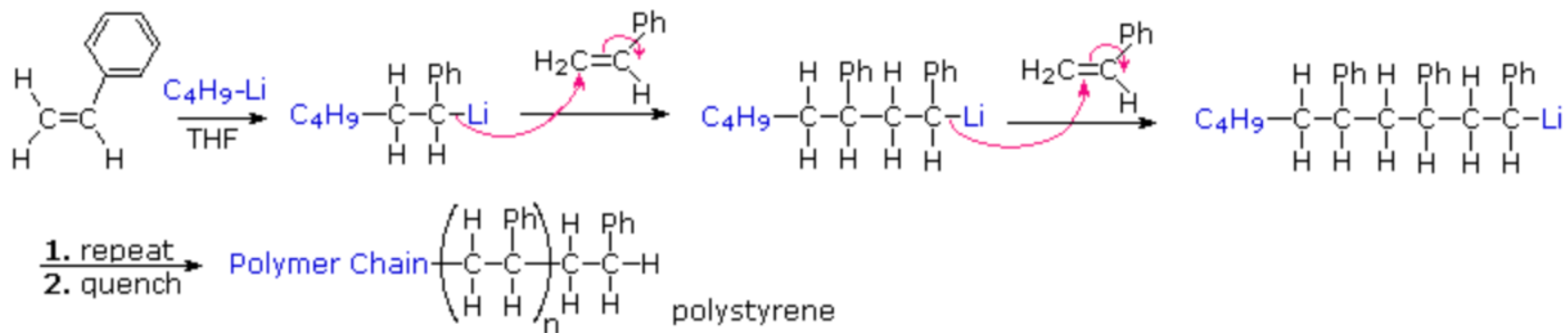


cross-linked homopolymer

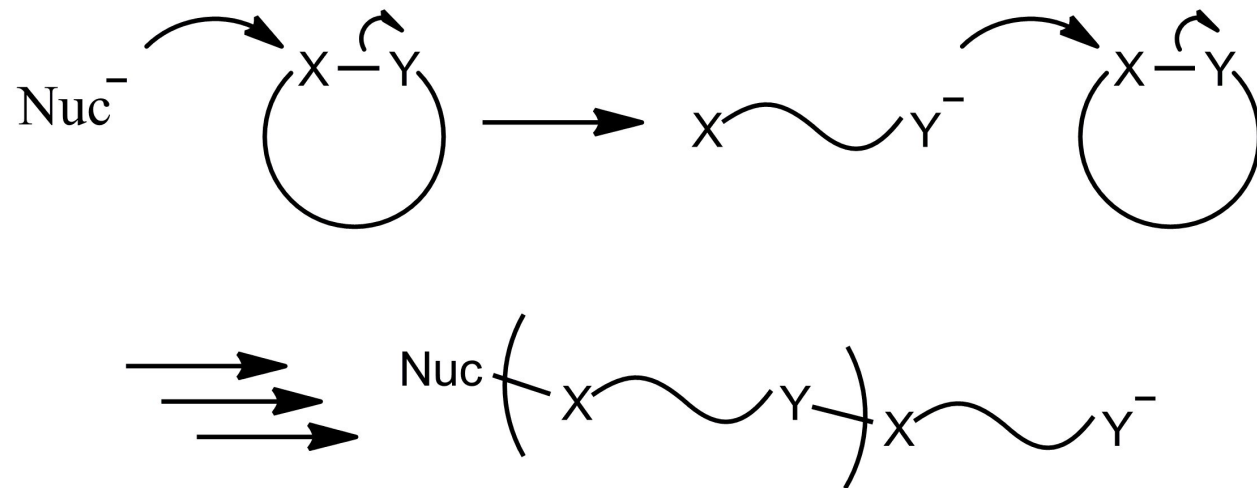
Cationic chain-growth polymerization



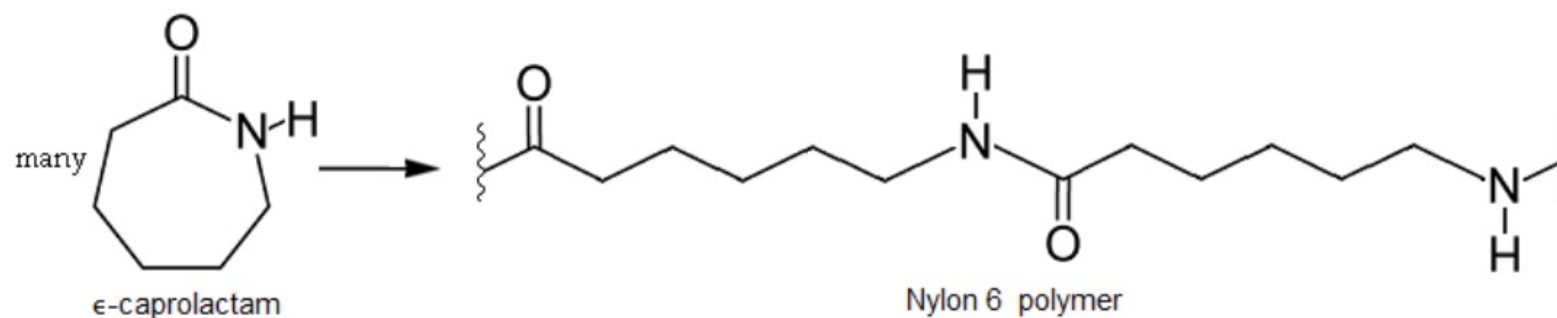
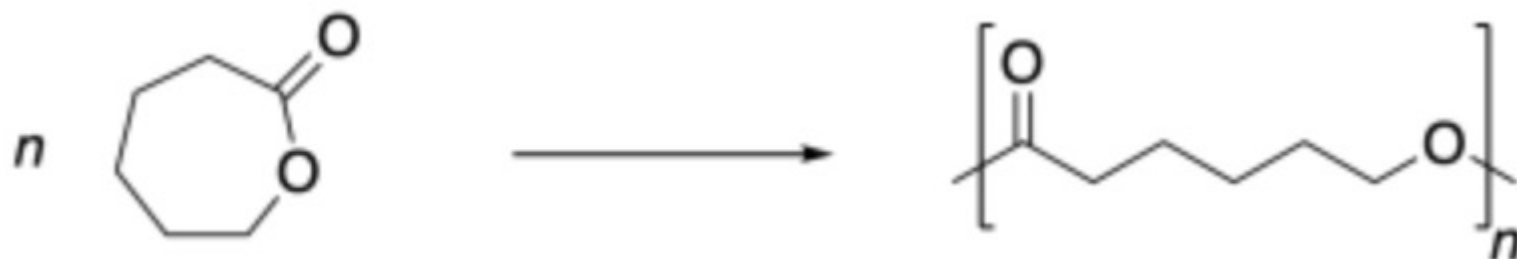
Anionic chain-growth polymerization



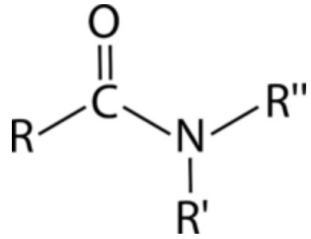
Ring opening polymerization



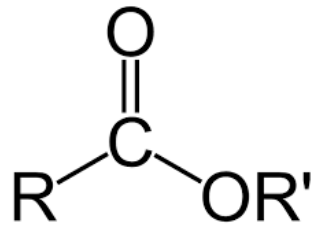
Lactones (cyclic esters) and lactams (cyclic amides)



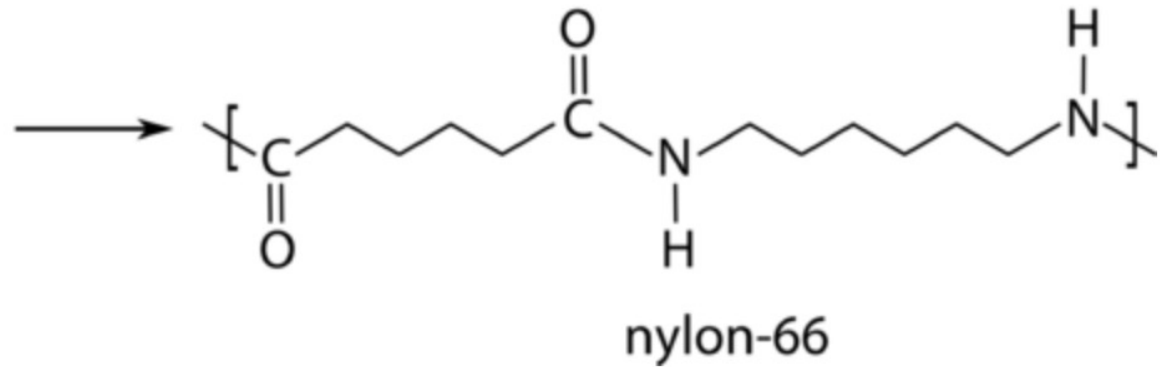
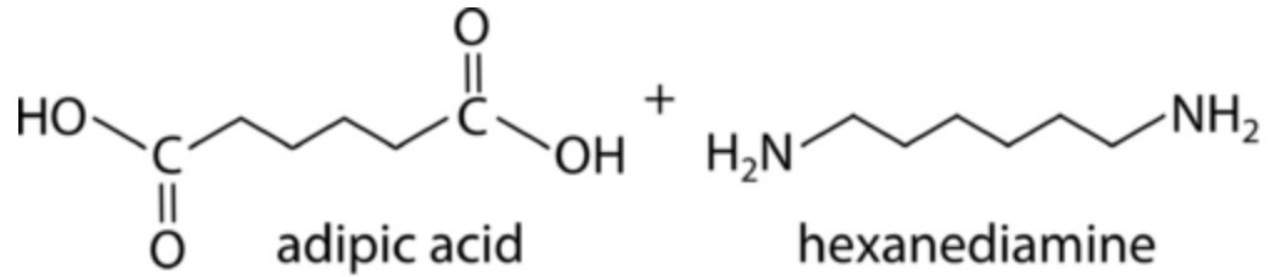
Condensation polymerization



Amide



Ester



Condensation polymerization

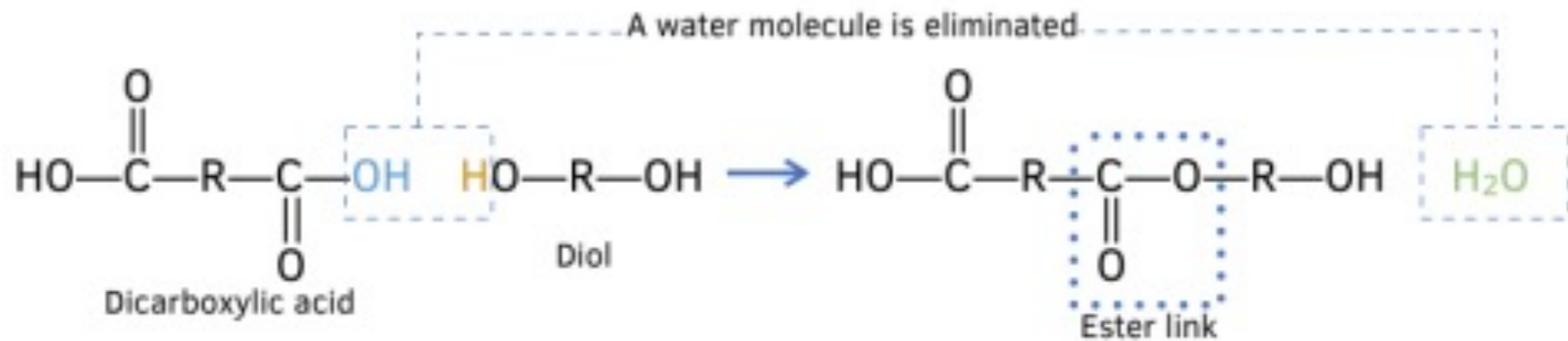
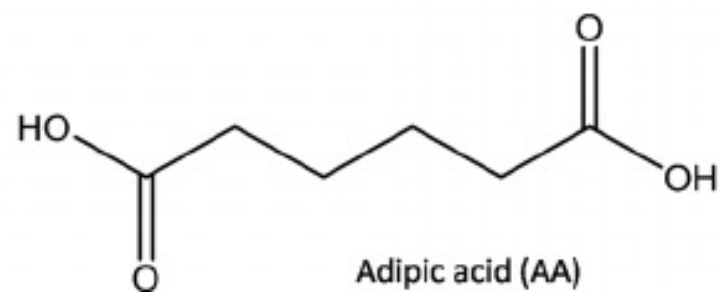
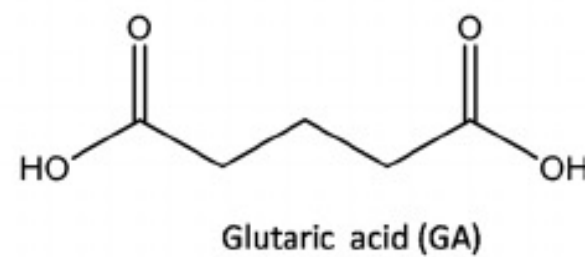
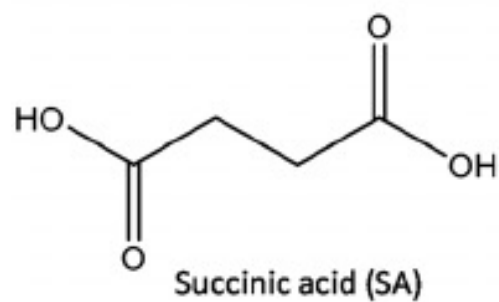
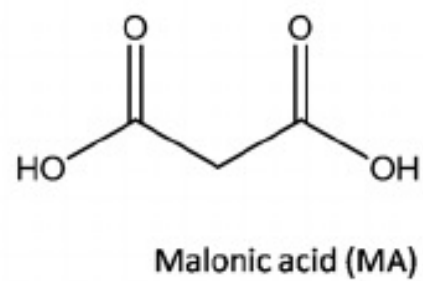
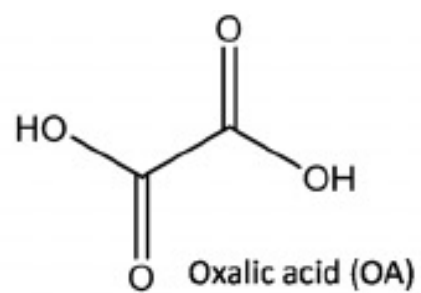
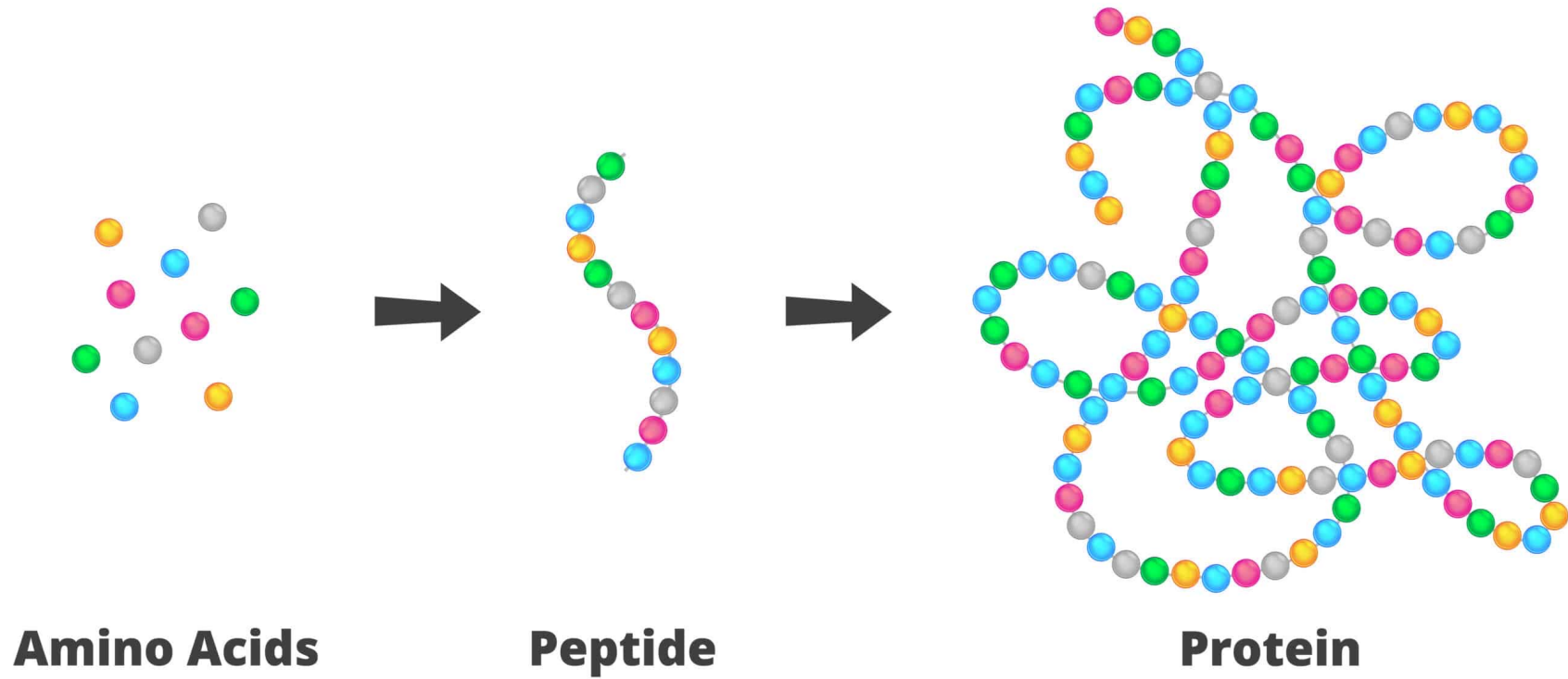


Fig 1. Condensation Reaction between a Dicarboxylic acid and a Diol to form a Polyester.

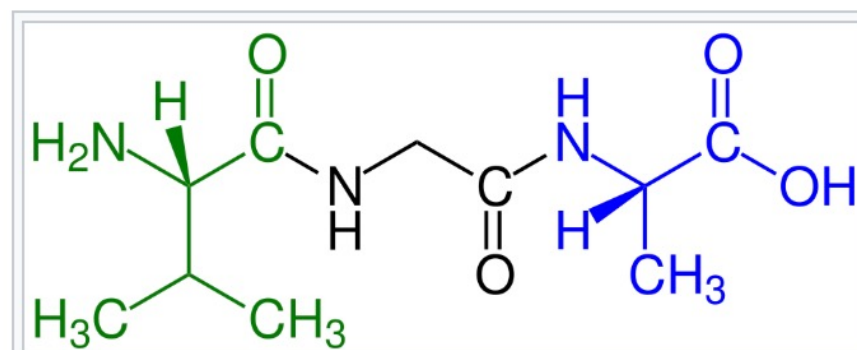


Nature is the best architect



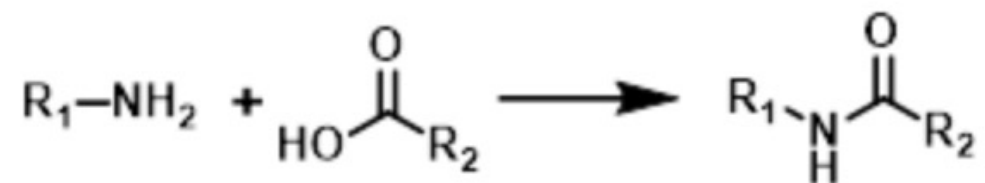
Bioconjugation Reactions: Peptide coupling

*Amides and ester bonds are
common in biological systems*



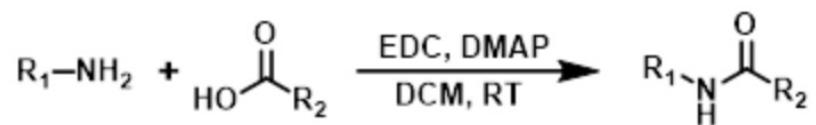
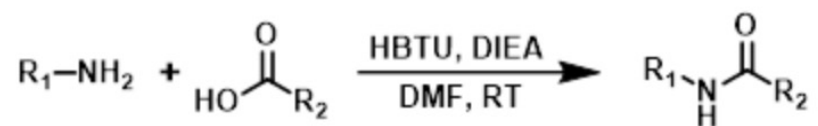
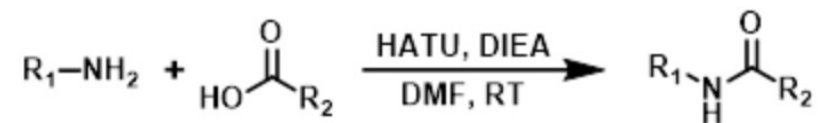
A tripeptide (example **Val-Gly-Ala**) with
green marked amino end (**L-valine**) and
blue marked carboxyl end (**L-alanine**)

Amide coupling reactions



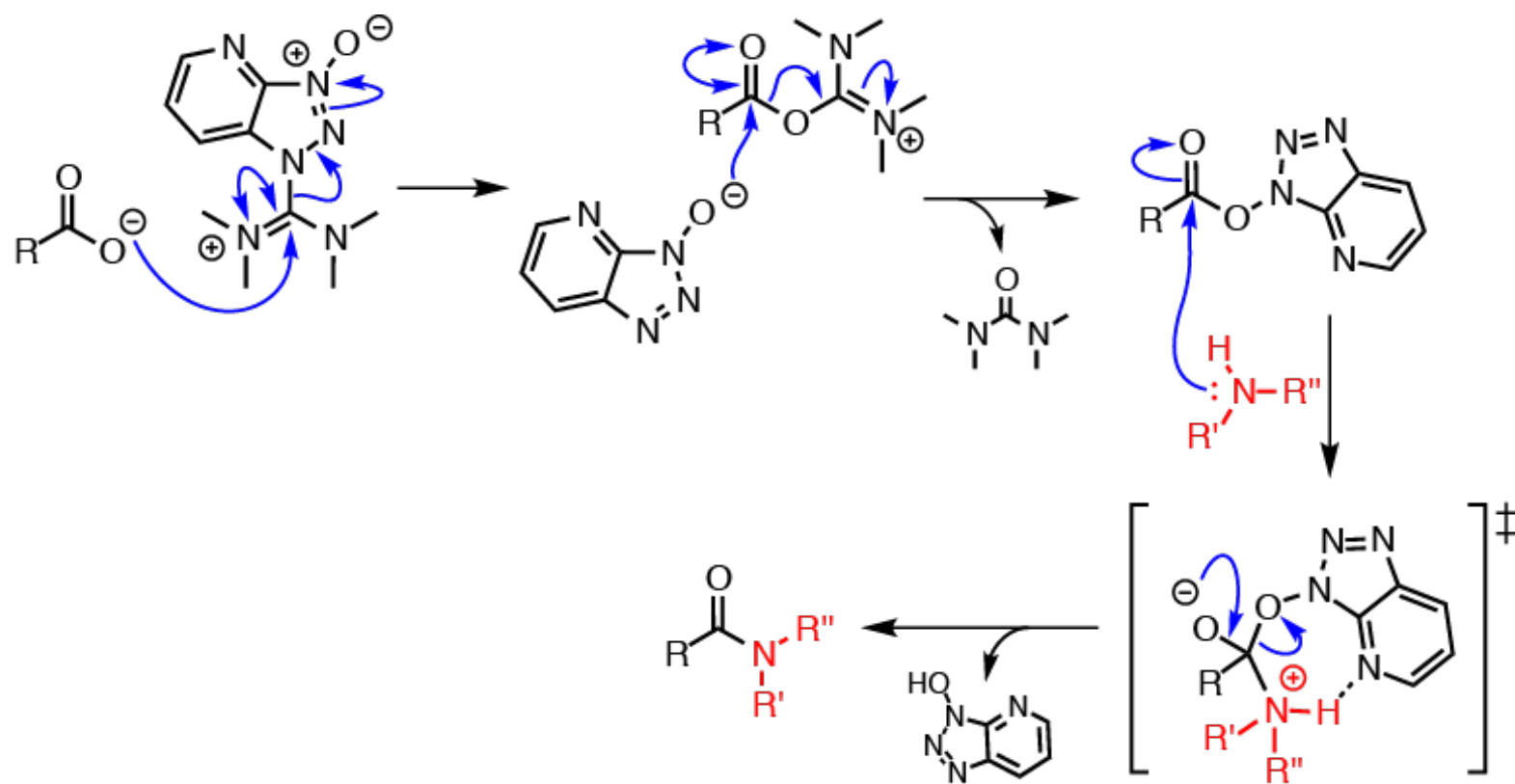
Acids need to be activated – weak electrophiles, poor leaving group

Peptide coupling reagents

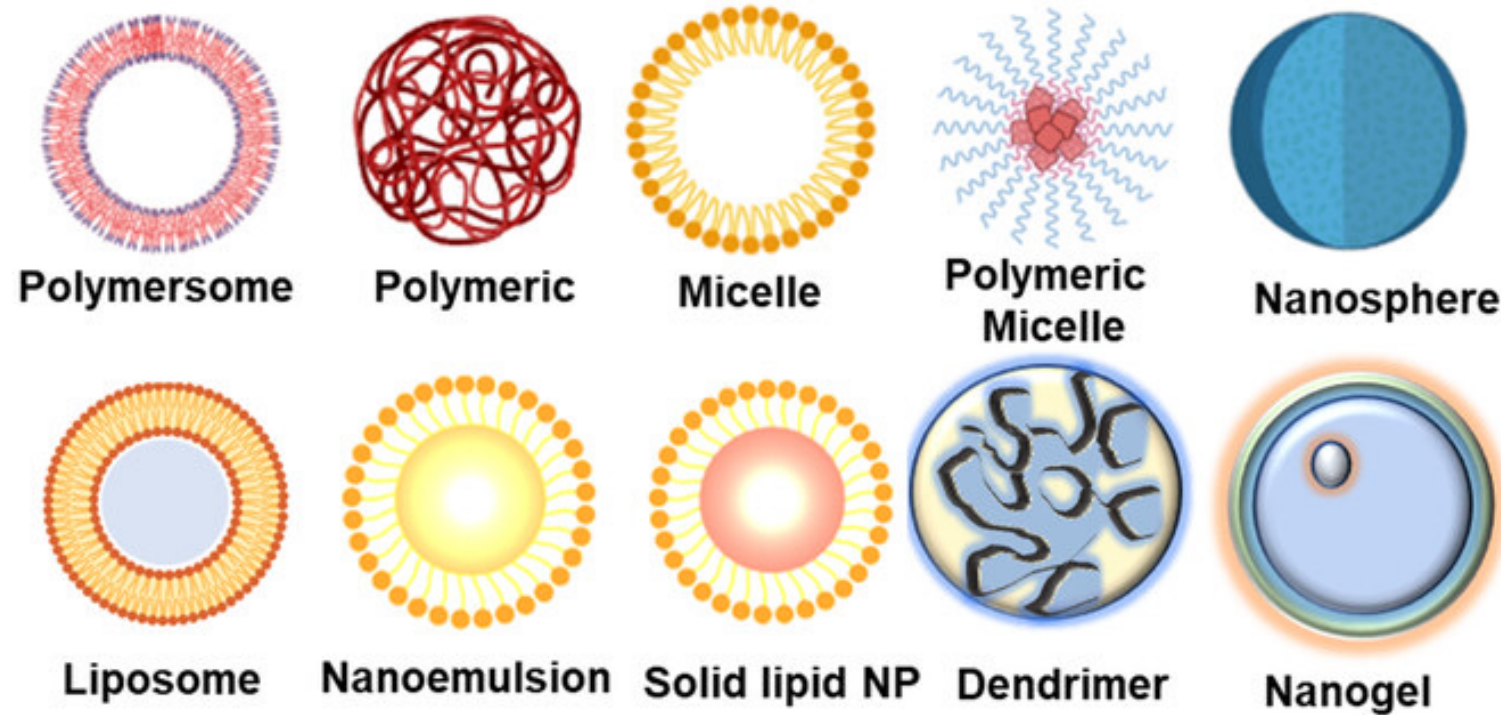


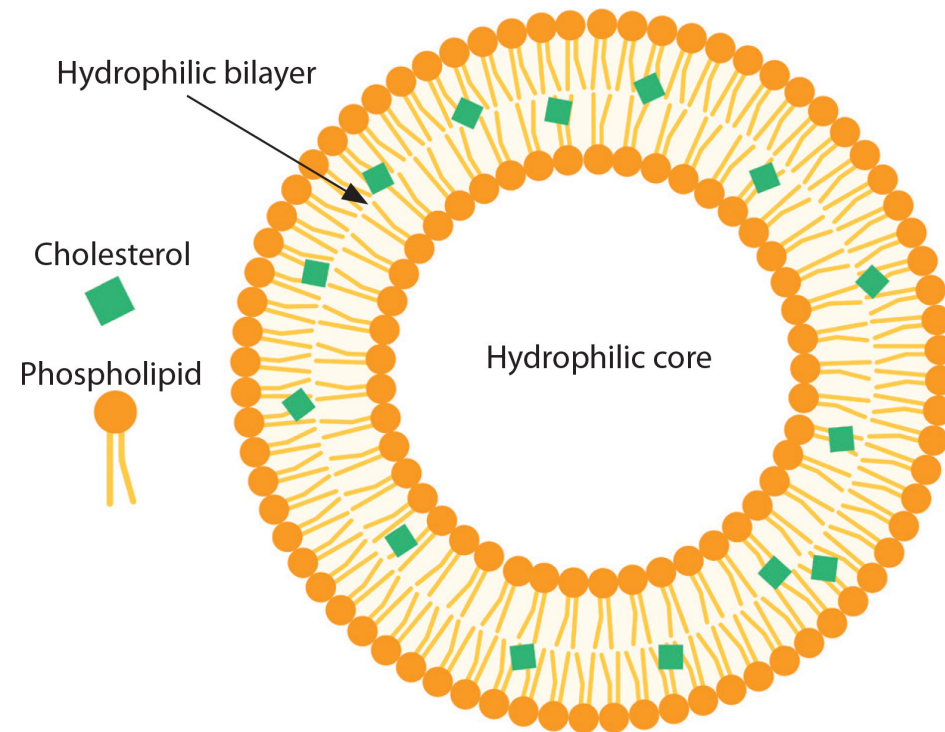
HATU, HBTU, DIEA, EDC, DMAP

Mechanism of activation



Polymers in drug delivery

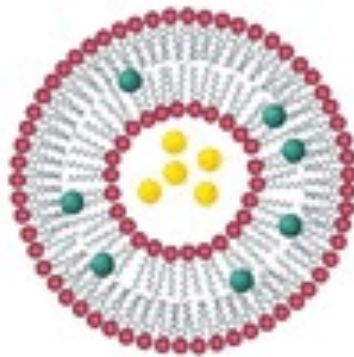




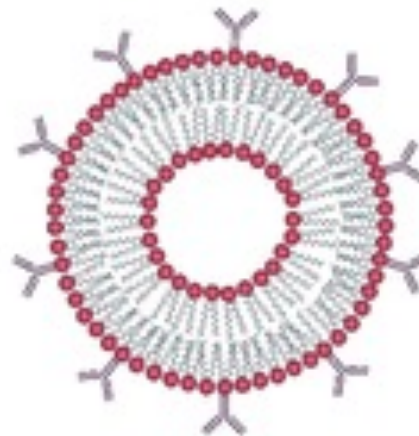
A lipid molecule
● - hydrophilic head group
- hydrophobic chains



B
● hydrophobic drug
● hydrophilic drug



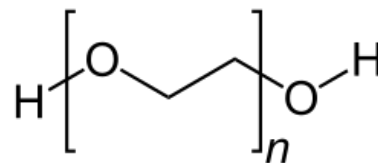
C
Y targeting ligand

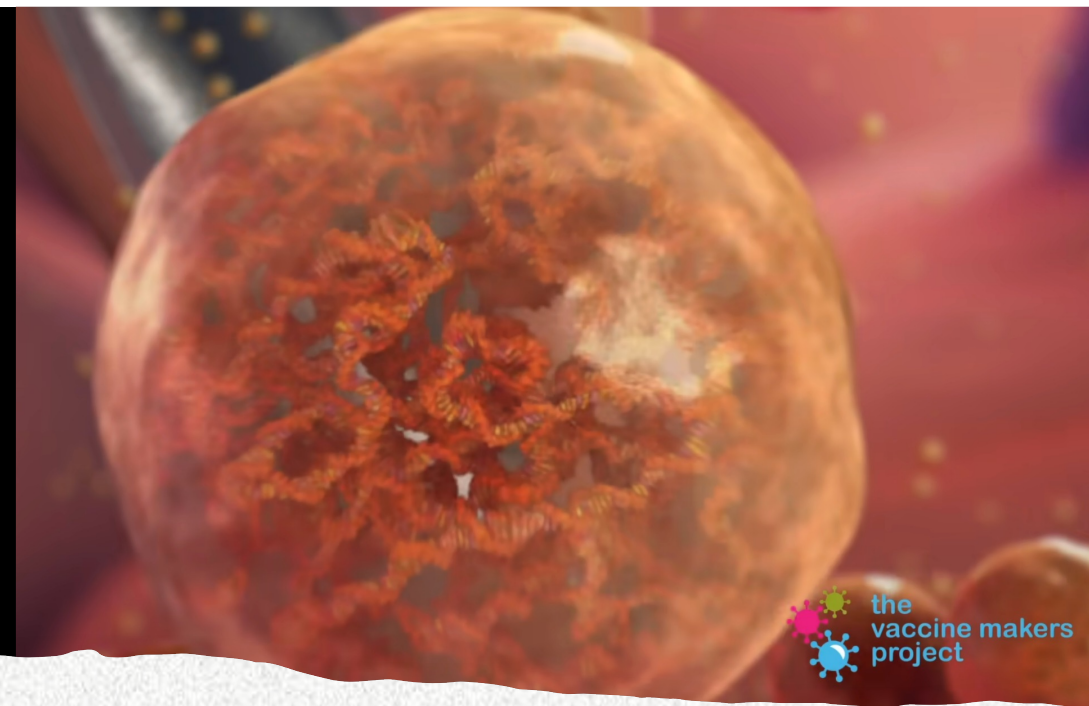
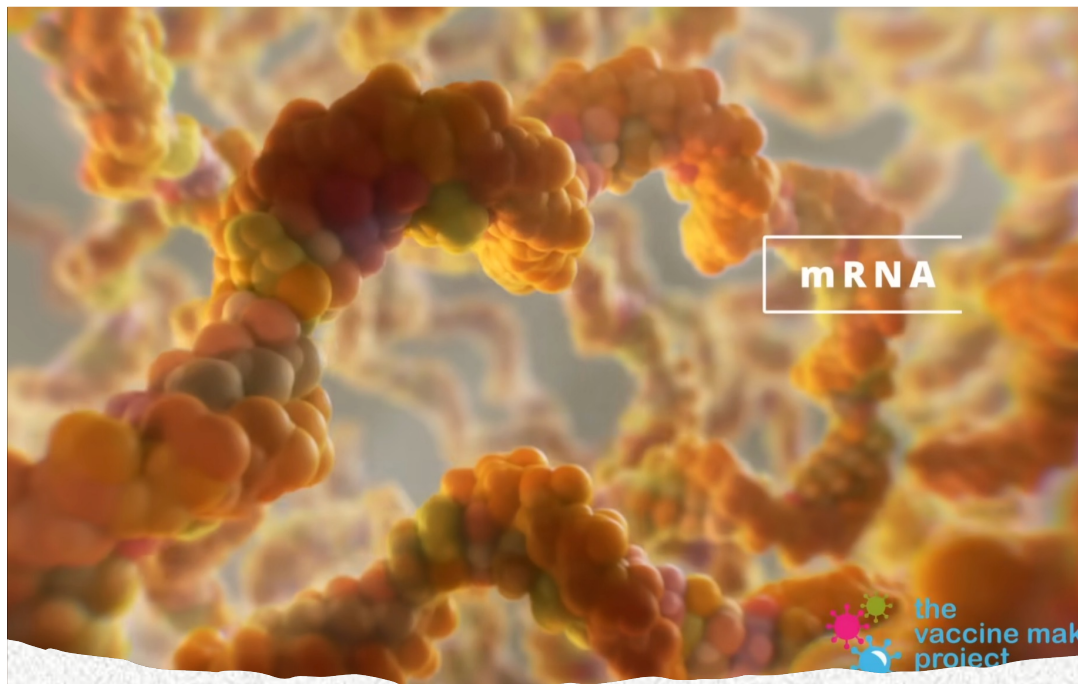


D
inert polymer such as PEG



PEG: Polyethylene glycol

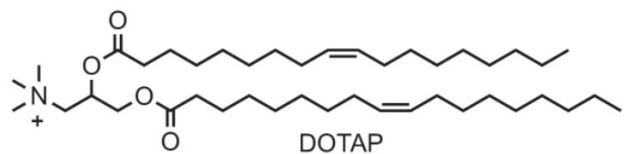
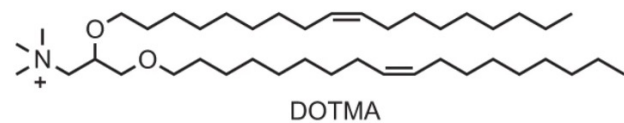




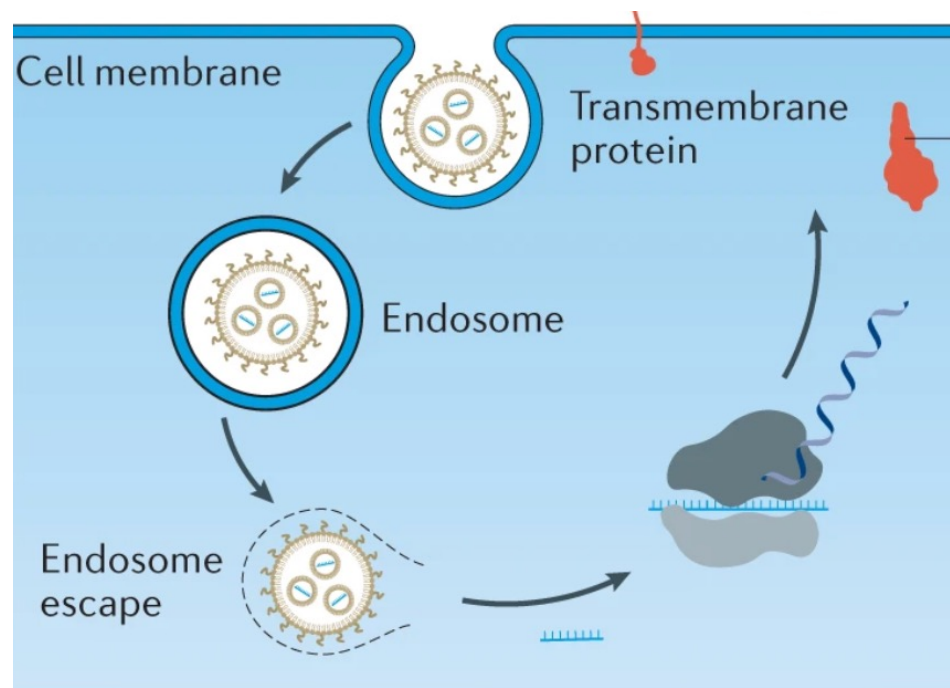
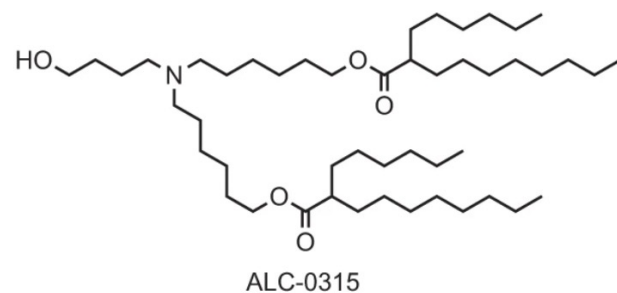
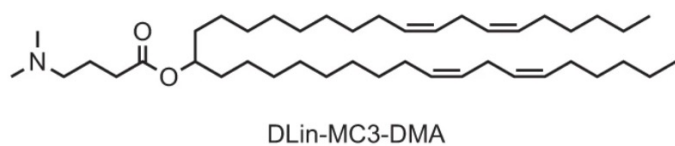
COVID VACCINES

Lipid based nanoparticles (LNPs)

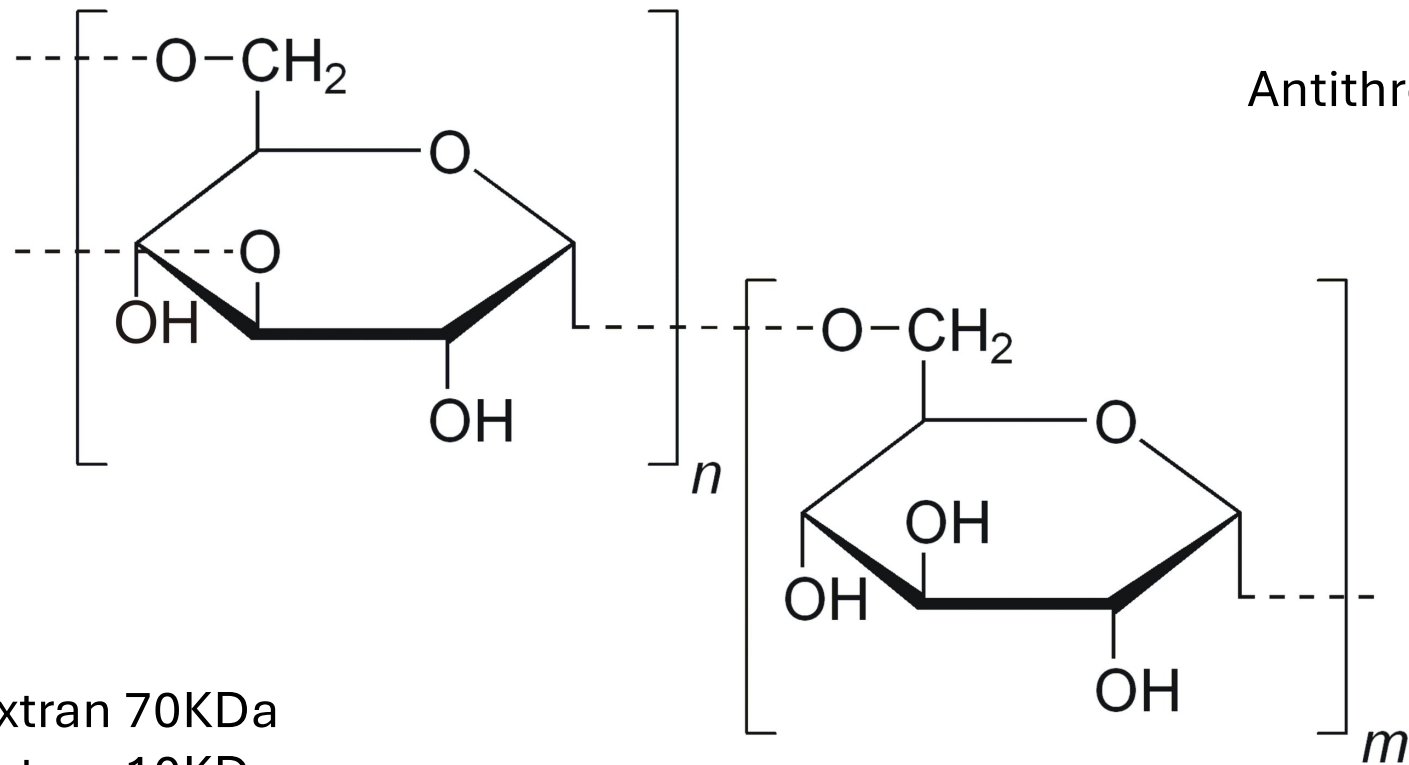
Cationic lipids



Ionizable lipids

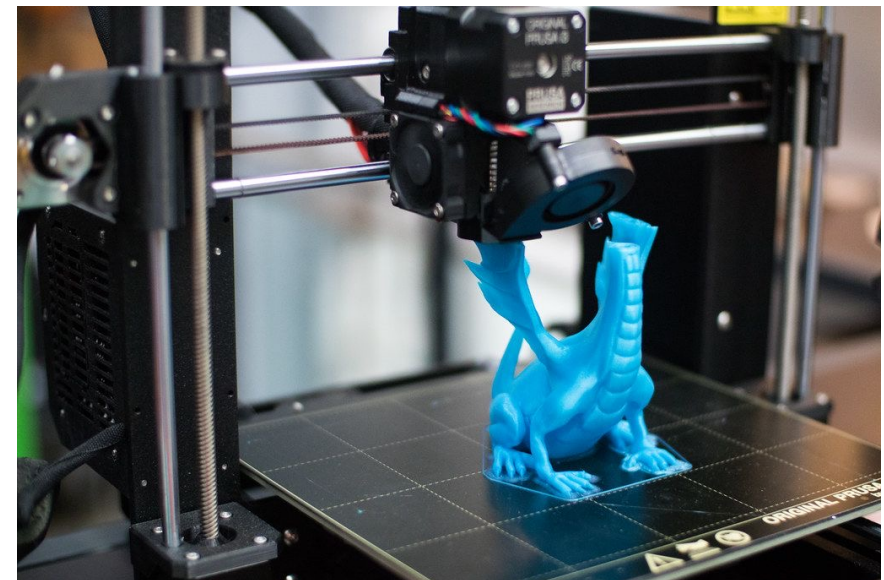
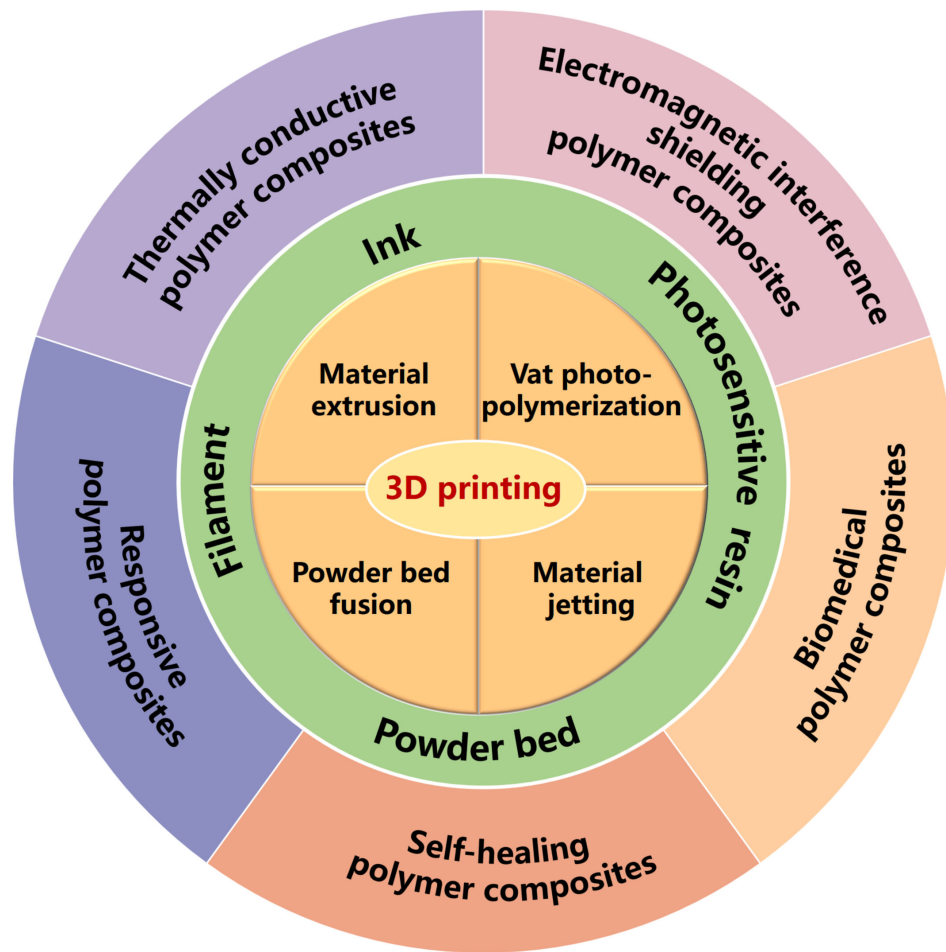


Dextran: Polysaccharides

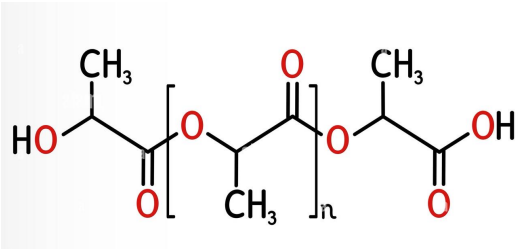


Dextran 70KDa

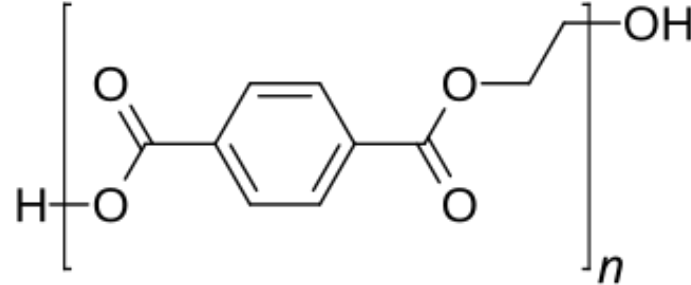
Dextran 10KDa



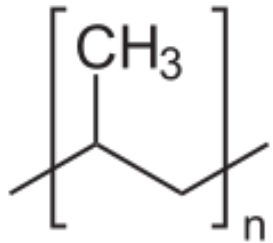
Polymers used in 3D printing



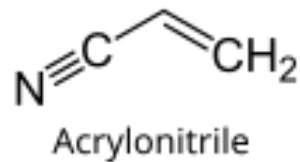
Polylactic Acid (PLA)



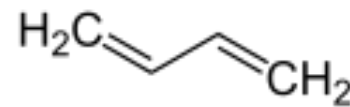
Polyethylene Terephthalate (PETG)



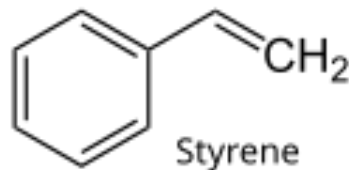
Polypropylene (PP)



Acrylonitrile



1,3-Butadiene



Styrene

Acrylonitrile butadiene
styrene (ABS)

